

Introduction

In 2024 and 2025, University of Maryland students in Professor Alam's and Professor Morawski's data science classes took a survey asking their opinions on multiple moral dilemmas. This report examines the trends in the responses. It was observed that there could be a correlation between the gender of the survey respondents and the individuals described in hypothetical scenarios, and the perceived "jerk" behavior. Responses were observed to not be affected by religion, political standing, or parents political standing. Graphs were used to compare the response distributions to all of the questions based on religion and political standing.

Background

The dataset comprises responses from students across four distinct collections, each containing demographic information. The respondents were asked their age, political and religious affiliations, gender, and school year. Each survey had 14 hypothetical questions that asked the respondents if the narrator was a jerk. Students rated the individuals in these scenarios as "not a jerk," "mildly a jerk," or "strongly a jerk." Two datasets presented scenarios with consistent gender roles, while the other two swapped the genders in the scenarios, allowing for a comparative analysis of how gender presentation influences judgment.

1 Data Issues and Cleaning

The initial data exploration revealed the following cleaning needs.

1.1 Empty Rows and Outliers: A completely empty row was identified and removed from the 2024 responses. Additionally, a graduate student aged over 50 was deemed an outlier and excluded from the analysis due to the dataset's focus on student populations. Missing entries were also present randomly throughout the dataset, which were dropped.

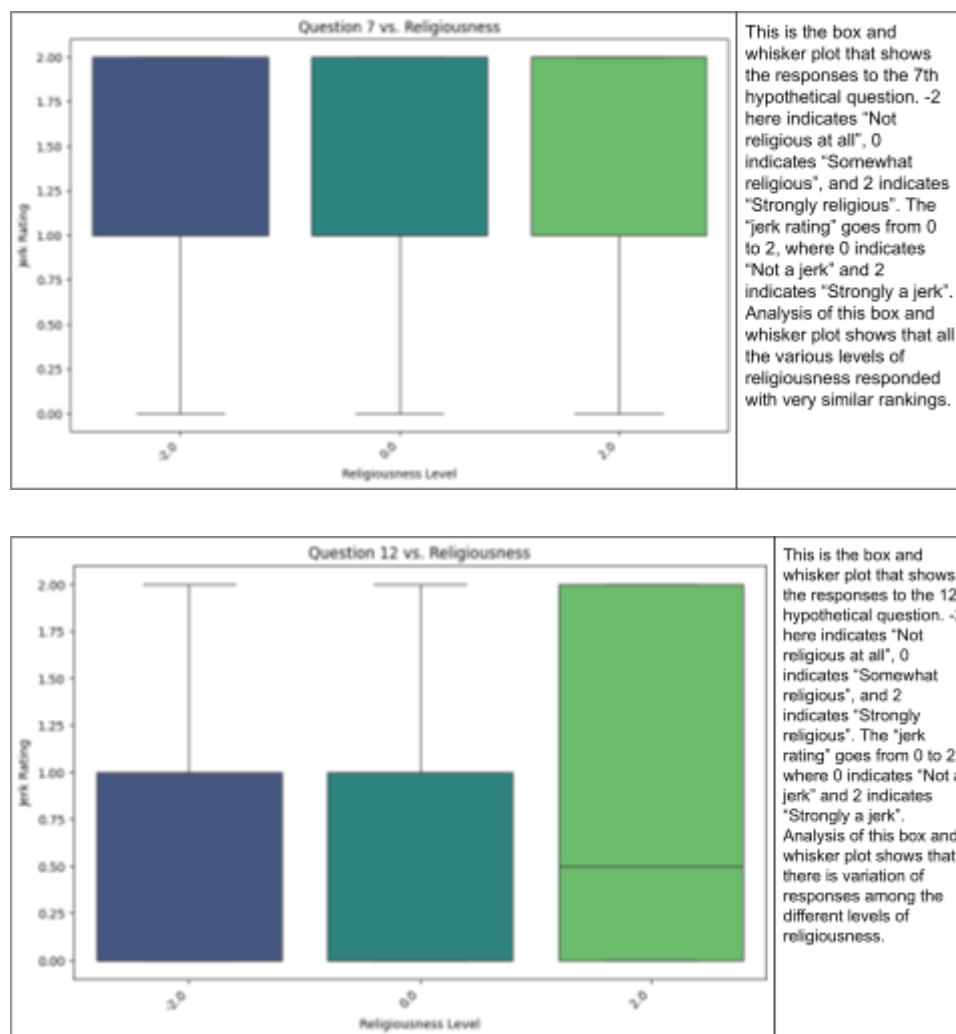
1.2 Data Consistency: Discrepancies in spelling (e.g., "famale" vs. "female," "sophmore" vs. "sophomore") were rectified using Google Colab. In addition, discrepancies in title formats (e.g., "Am I a jerk" vs. "am i jerk," newlines) were standardized. The column name that asked about religiousness was different in two of the data sets, asking for religiousness / spirituality instead of religiousness, so that was changed to only say religiousness.

1.3 Column Alignment: A "compassionate" column present in one dataset (Fardina's dataset) was removed to ensure uniformity across all datasets, as this column was not present in the other datasets. Removing this dataset made it so that all of the hypothetical question columns and all of the demographic question columns were aligned.

1.4 Categorical Data: Hypothetical question responses such as "strongly a jerk", "mildly a jerk", and "not a jerk" were mapped to 2, 1, 0 respectively. Responses that asked to rank religious, and spiritual standing were also mapped to numbers similarly to conserve standardization.

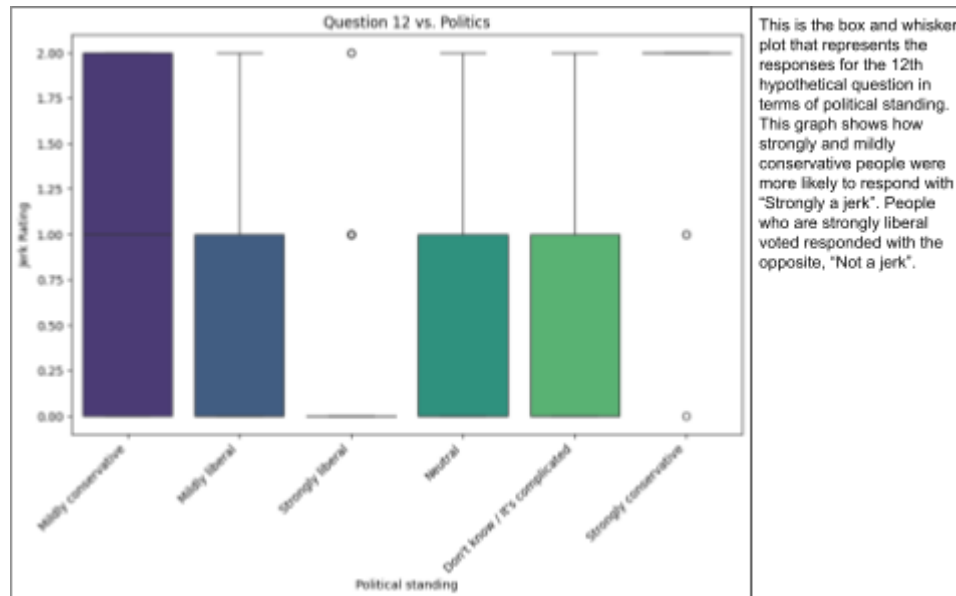
2 Findings

2.1 Question 1: Does religion affect the responses to the hypothetical questions? A chi-squared test was performed to see if there was any statistical significance between these values. A Bonferroni correction was also applied because we are conducting multiple tests. box plots were also made for religion and judgment of jerk behavior for each hypothetical question. The chi-squared test with the Bonferroni correction revealed that the responses to question 7 and 12 might be correlated to the religion of the respondent- the p-values were 0.001 and 0.00 respectively. The analysis of the chart for the 7th hypothetical question indicated no discernible correlation between religious affiliation and the judgment of jerk behavior. This could be due to the fact that there might be subtle patterns in the data that are captured by the chi-square test but not easily visible in the graph. The analysis of the chart for the 12th hypothetical question showed that people that were strongly religious responded more with “Strongly a jerk” or “Mildly a jerk” than people who weren’t strongly religious.



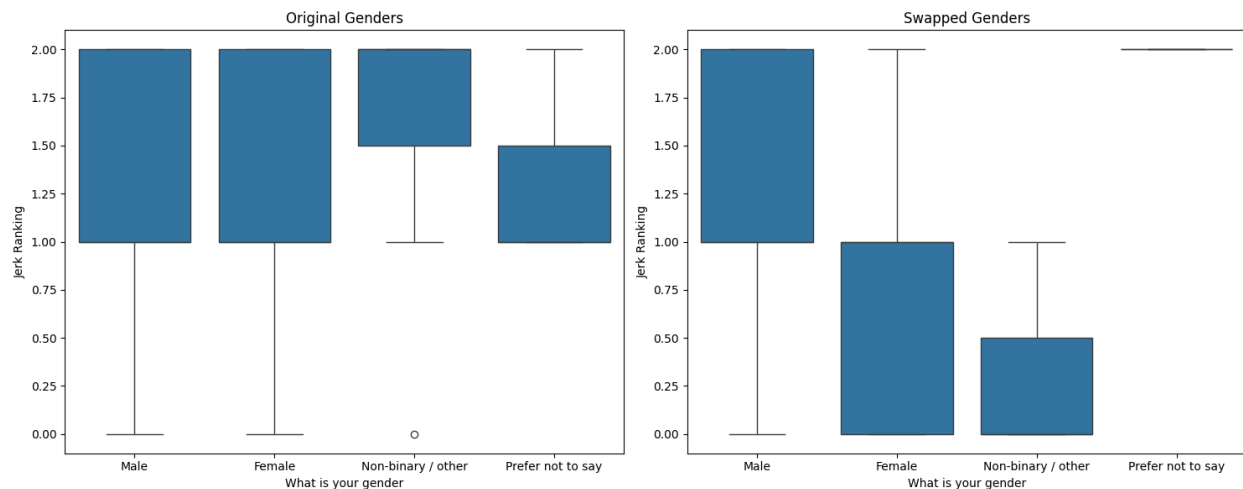
2.2 Question 2: Does Political standing affect the responses to the hypothetical questions? Similar to the previous question, a Chi-square test was performed with a Bonferroni correction to

check for statistical significance. Box and whisker plots were also made for each of the hypothetical questions. Through the Chi-square tests, it was revealed that the responses to question 12 might be related to the respondents political standing. Question 12 had a p-value of 0.00 and the box and whisker plot shows that people that are strongly or mildly conservative responded more with “Strongly a jerk” or “Mildly a jerk” than people that were mildly liberal, neutral, or didn’t know. People who were strongly liberal responded mostly with “Not a jerk”. This suggests that there could be a correlation between political standing and responses in certain contexts.



2.3 Question 3: Does gender in the question and gender of the respondents affect the responses? Box and whisker plots revealed that while male and female respondents generally exhibited similar response patterns, non-binary individuals and those who preferred not to specify their gender showed greater variability in their judgments. Sometimes people that were non-binary or preferred not to reveal their gender had no responses to questions, other times they would be very different from both the male and female responses. Comparative box and whisker plots, analyzing scenarios with original and swapped genders, demonstrated that the distribution of responses remained largely consistent, with one notable exception: Question 7, concerning familial obligations, displayed a significant gender-based response variation. Specifically, when the scenario involved sisters’ kids, female and non-binary respondents were less likely to perceive the individual as a jerk. Conversely, when the scenario involved brothers’ kids, all respondent groups, including males, showed a higher propensity to judge the individual as a jerk. The males’ responses stayed consistent for both hypothetical situations. A two-tailed two-sample t-test specifically for the 7th hypothetical question yielded a p-value of 0.062, exceeding the significance level of 0.05. This result led to the initial conclusion of failing to reject the null hypothesis, suggesting no significant correlation between respondent gender and scenario gender in judging jerk behavior. A subsequent one-tailed two-sample t-test, specifically examining whether jerk responses decreased when the scenario gender was swapped to involve sisters’ kids, produced a p-value of 0.03. This result, below the 0.05 significance level,

led to the rejection of the null hypothesis, indicating a statistically significant correlation. This suggests that gender may be correlated to responses in certain situations. Below is a visualization that shows the responses based on what gender is in the question and the gender that is being asked in the hypothetical. Original genders represents the graph of the question involving brothers' kids, swapped genders represents the graph of the question involving the sisters' kids



Overall, Box and whisker plots were employed to compare response distributions across different respondent gender groups and to analyze the impact of scenario gender swapping. Chi - square tests and t-tests were used to check for statistical significance.

Conclusion

In conclusion, this report examined survey responses from University of Maryland students regarding their perceptions of "jerk" behavior in hypothetical scenarios, focusing on the influence of demographic factors. While initial explorations suggested potential correlations between religion and political standing with responses to certain questions, specifically questions 7 and 12, further analysis revealed that the observed patterns were subtle and often not visually discernible in box plots. However, statistically significant results from Chi-square tests indicated that religion and political standing may play a role in shaping responses to these specific scenarios.

The most notable finding emerged from the analysis of gender dynamics. Although overall response patterns were largely consistent across male and female respondents, significant variations were observed among non-binary individuals and those who preferred not to specify their gender. Moreover, a one-tailed t-test revealed a statistically significant correlation between the gender presented in the scenario and the respondents' judgments, particularly in the context of familial obligations. Specifically, respondents were less likely to perceive an individual as a jerk when the scenario involved sisters' kids compared to brothers' kids. This suggests that gender roles and societal expectations may influence moral judgments in specific contexts.

Ultimately, while religion and political standing showed some statistical correlation with responses, the most prominent factor influencing judgments appeared to be the interplay between respondent gender and the gender presented within the hypothetical scenarios. This highlights the complexity of moral perception and the subtle ways in which demographic factors can shape our judgments. Future research could explore these findings in greater depth, potentially using larger datasets and more nuanced scenario designs to further elucidate the relationships between gender, religion, political standing, and moral judgments